VIETNAM NATIONAL UNIVERSITY, HANOI

UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Hoang Duc Bach

DESIGN AND IMPLEMENTATION OF AN INTENT-BASED META DEX AGGREGATOR WITH COINCIDENCE-OF-WANTS MATCHING ON SUI BLOCKCHAIN

BACHELOR THESIS

Major: Information Technology

HA NOI - 2026

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Acknowledgments

I would like to express my sincere gratitude to my supervisor, Dr. Nguyen Thi Hau, for her guidance, valuable feedback, and continuous support throughout this research. Her insights and suggestions greatly helped improve the quality of this thesis.

I also gratefully acknowledge the support of the Sui and Deepbook teams for their generous provision of testnet token funding, which made the practical implementation of this research possible. I extend equal thanks to the broader Sui community for their open and collaborative spirit — their technical discussions, shared resources, and willingness to engage with emerging research have enriched this work in ways that extend well beyond what any single reference could capture.

I am equally thankful to my family, whose steady encouragement and unwavering belief in me sustained me through the most demanding stages of this work.

This thesis is, in many ways, a reflection of the collective generosity of those who supported me — academically and personally. To each of them, I am sincerely grateful.

Declaration

I, Hoang Duc Bach, declare that this thesis is my own original work, composed independently under the supervision of Dr. Nguyen Thi Hau. It has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution.

All sources consulted in the preparation of this thesis have been fully acknowledged and cited in accordance with academic convention. I accept full responsibility for the integrity of this work and for any consequence arising from a failure to meet this declaration, in accordance with the regulations of the University of Engineering and Technology, Vietnam National University.

Undergraduate student,

Hoang Duc Bach

Abstract

As the decentralized finance (DeFi) landscape continues to evolve, slippage and Maximal Extractable Value (MEV) remain significant hurdles for users on Ethereum and EVM-compatible chains. While CoW DEX solutions have successfully mitigated these issues in the EVM space, the Sui network presents a unique environment where its Object-Centric model inherently provides robust protection against certain forms of MEV. Nevertheless, slippage continues to be a persistent challenge for decentralized exchanges (DEXs) on the Sui network.

This thesis focuses on the analysis, architectural design, and development of a CoW DEX on Sui, leveraging the network's object-oriented paradigm and inspired by the core architecture of the Ethereum CoW Protocol. The research proposes a novel auction mechanism incorporating a Commit-Reveal phase and utilizes the Hot Potato Pattern to ensure atomicity, fairness, and the prevention of solver fraud. The system aims to establish an intent-based DEX aggregator that delivers optimal pricing and zero slippage for coincident orders (Coincidence of Wants), while effectively harnessing Sui's core liquidity layers, specifically Deepbook. This work contributes a scalable and secure trading framework that enhances transparency and capital efficiency within the Sui ecosystem.

Keywords: *Blockchain, Decentralized Finance, Coincidence of Wants, Slippage, Maximal Extractable Value, Intent-based, Object-Centric, Sui, Ethereum, Deepbook, CoW Protocol.*

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Chapter 1

The title of chapter one

There's something to be said for having a good opening line. Morbi commodo, ipsum sed pharetra gravida, orci $x = 1/\alpha$ magna rhoncus neque, id pulvinar odio lorem non turpis [1, 2]. Nullam sit amet enim. Suspendisse id velit vitae ligula volutpat condimentum. Aliquam erat volutpat. Sed quis velit. Nulla facilisi. Nulla libero. Vivamus pharetra posuere sapien. Nam consectetuer. Sed aliquam, nunc eget euismod ullamcorper, lectus nunc ullamcorper orci, fermentum bibendum enim nibh eget ipsum. Donec porttitor ligula eu dolor. Maecenas vitae nulla consequat libero cursus venenatis. Nam magna enim, accumsan eu, blandit sed, blandit a, eros.

$$\zeta = \frac{1039}{\pi}$$

For an example of a full page figure, see Fig. 1.2.

1.1 This is section one

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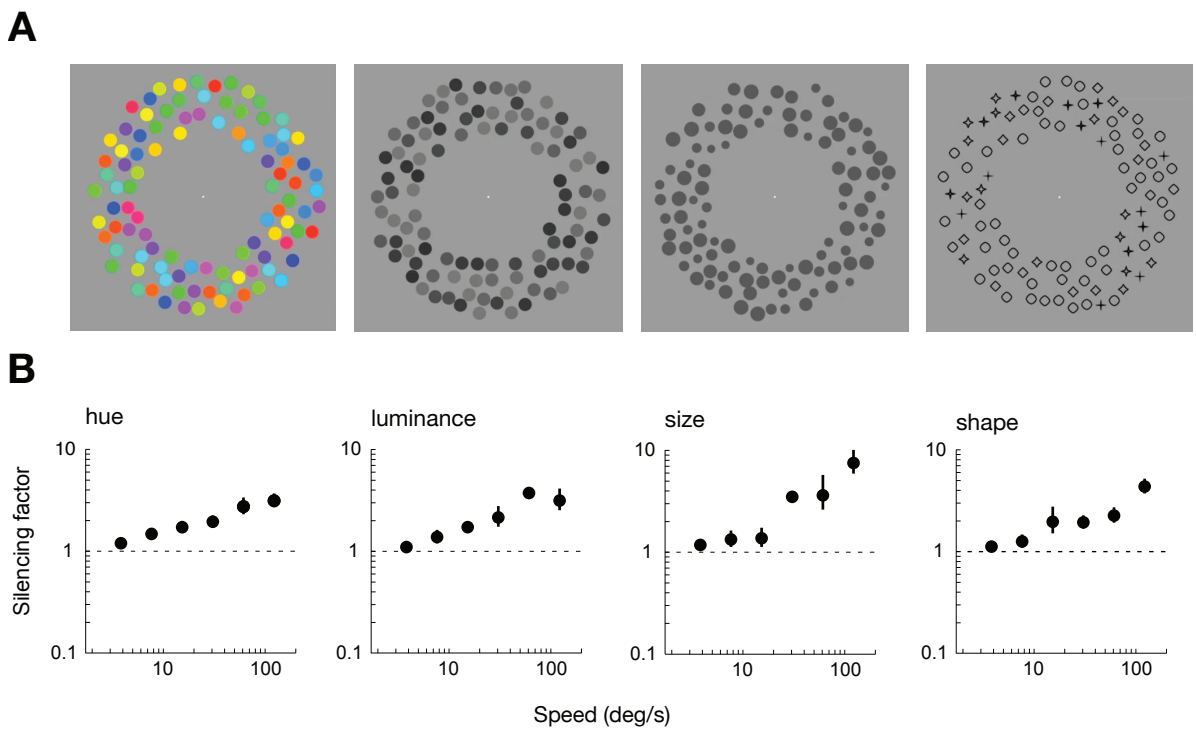

Figure 1.1: This is a figure that floats inline and here is its caption.

language. There is no need for special content, but the length of words should match the language.

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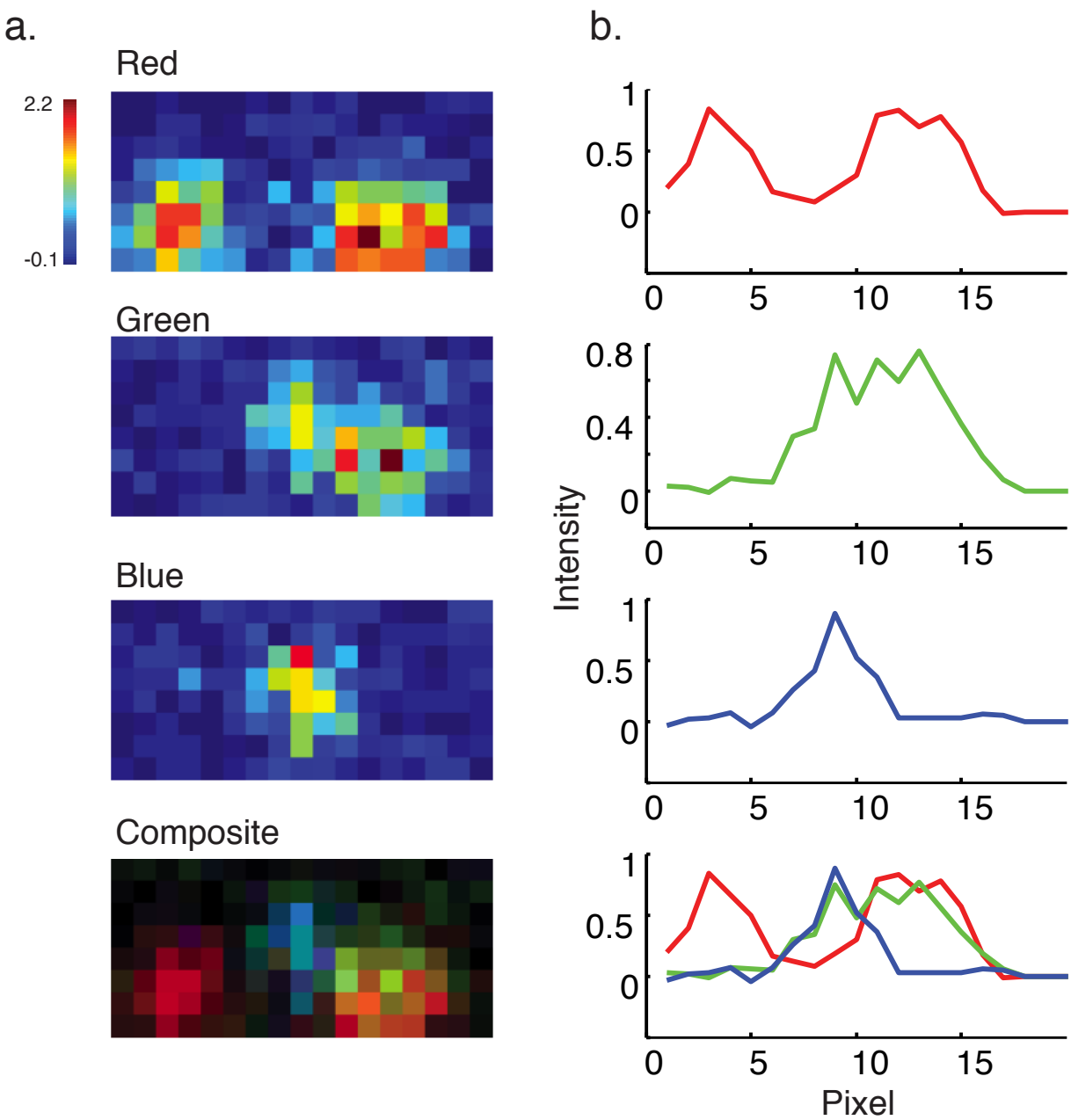

Figure 1.2: This is a full page figure using the FPfigure command. It takes up the whole page and the caption appears on the preceding page. Its useful for large figures. Harvard's rules about full page figures are tricky, but you don't have to worry about it because we took care of it for you. For example, the full figure is supposed to have a title in the same style as the caption but without the actual caption. The caption is supposed to appear alone on the preceding page with no other text. You do't have to worry about any of that. We have modified the fltpage package to make it work. This is a lengthy caption and it clearly would not fit on the same page as the figure. Note that you should only use the FPfigure command in instances where the figure really is too large. If the figure is small enough to fit by the caption than it does not produce the desired effect. Good luck with your thesis. I have to keep writing this to make the caption really long. LaTex is a lot of fun. You will enjoy working with it. Good luck on your post doctoral life! I am looking forward to mine.

about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Firstname lastname

Chapter 2

The title of chapter two

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- First item in a list

Table 2.1: Network constants of the energy model.

Parameter	Value	Unit
ϵ_{elec}	50	nJ/bit
ϵ_{fs}	10	$pJ/bit/m^2$
ϵ_{mp}	0.0013	$pJ/bit/m^4$

- Second item in a list
- Third item in a list
- Fourth item in a list
- Fifth item in a list

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Chapter 3

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3.1 This is section one

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$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n\to\infty} \sum_{k=0}^{n} a_0 q^k = \lim_{n\to\infty} a_0 \frac{1-q^{n+1}}{1-q} = \frac{a_0}{1-q}$$

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$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

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Appendix A

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Appendix B

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